

Experience Summary:

- About 16 years of experience in leading applications and driver development with end-to-end responsibilities using GoLang, Python3, C, C++ in Distributed Systems and Client/Server low latency network applications.
- Extensive experience with multiple technologies – **GoLang with gRPC Protocol Buffers and Gorilla mux, Python3, C++.**
- Cognitive about designing, deploying with Terraform and operating highly available, scalable and fault tolerant systems using Amazon Web Services (AWS).
- Experience in AWS core technologies, EC2 (Elastic Compute), S3 (Simple storage service), DynamoDB.
- Experience in domains such as Financial Quantitative modeling technologies, Embedded Logix Controllers, High availability Clustering, Telecom, Infrastructure management and Utility Software domains.
- Experience in inter process communication (Pipes and shared memory), multithreaded applications (POSIX threads) in C/C++ using IPC Synchronization mechanisms (IBM Message queues, semaphore, mutex etc.), Socket Programming and developing applications over TCP-IP/UDP protocol and SNMP Protocol.
- Experienced with event-driven and scheduled AWS Lambda functions to trigger various AWS resources using CloudWatch events.
- Experience in Docker configurations for multi-node API Gateway infrastructure.
- Managed Docker containerization using Kubernetes.
- Displayed and proved capability to learn any new technology and work on it in a matter of few days.
- Experience in Perl, Python & Shell scripting.
- Experience in production supports and meeting customer requirements in different level.
- Fluency with C++ Data structures, algorithms, Exception handling, Memory Management and debugging using gdb/dbx and running memory mismanagement detector Valgrind.
- Experience in Handling customer calls and providing support for production and development issues, bug fixing, support and maintenance.
- Fluency with Algorithms, STL, data structures Exception handling, Memory Management, debugging using gdb and object-oriented design.
- Possess good knowledge in AWS and creating and launching EC2 instances using AMI's of AMD64 Linux.
- Expertise in working with Terraform as infrastructure as code for automating deployments and reducing human error.
- Experience in working with Design Patterns.
- Experience in Agile/Scrum and Waterfall models.
- Experience in GIT, GITLAB, CMVC, VSS, Jenkins CI/CD and Jira (Issue and Project tracking software)
- Effective problem solving and troubleshooting abilities.
- Experience in proof of concepts preparation on emerging technology or domain.
- Solid Planning and Organizational Skills in all aspects. Well-disciplined with proven ability to manage multiple assignments effectively under extreme pressure while meeting tight Schedule deadlines.

EDUCATIONAL QUALIFICATION:

- Bachelor of Technology in EEE in APRIL 2005 - Nagarjuna University, Guntur, India
- Certification in Embedded Systems, India

TECHNICAL SKILLS:

Programming

GoLang, Python3, and C, C++.

Scripting Languages

Shell scripting, Perl, Python Programming

Operating Systems

MacOS, IBM-AIX, HP-UX, LINUX (RHEL 6*, RHEL 7*, SLES 12 SP3, UBUNTU 17.10)

WORK EXPERIENCE:

1) Employer: V-Soft Inc - September 2024 to Till Date

Client: Capital One, Texas (Contract)

Senior Golang Developer

Project: Business Core

Description:

Business Core is the single source of truth for business client data. It delivers a holistic view of client organizational structures, relationship connections, and comprehensive client and account information, all in one central platform.

Responsibilities:

- Designed and implemented high-performance **gRPC** services in **Golang** that build a unified, commercial client repository that stores accurate data with strong controls.
- Updated features in cone-api-service, a backend component that runs **GraphQL** server on a **AWS fargate** container that serves as the Backend for the API.
- Work on **AWS lambda** that handles requests to the Business Entity API, RelatedParty Relationship API and Accounts API.
- Work on **SQS** Queues that triggers **AWS lambda** to create Business core audits in AWS DynamoDB which will provide an audit trail for changes made to Business Entities and Related Parties.
- Implemented **API performance testing** for Go microservices using **Hercules**, benchmarking throughput, latency, and concurrency limits across production-like environments.
- Designed and executed load, and stress tests in Hercules to identify performance bottlenecks in high-traffic REST APIs built in Go.
- Integrated **OpenTelemetry (OTel)** instrumentation into Go microservices to capture distributed traces, metrics, and logs for real-time performance visibility.
- Configured **OTel SDK and exporters** to send telemetry data from Go APIs to **New Relic**, enabling unified observability across services.
- Implemented **custom spans and attributes** in critical API paths to trace latency, database queries, and external AWS service calls (S3, SQS, DynamoDB).
- Implemented **SNS topic publishing** from AWS lambda, publishing in Go with delivery status tracking to ensure successful notification delivery to subscribers.
- Configured **CI/CD pipelines in Jenkins pipelines** for build, test, performance benchmarking (Hercules), and deployment of Go applications.
- Implemented robust unit tests using Go's **testing** package and end-to-end acceptance tests with **Godog** to ensure reliable and fully validated Go microservices.

2) Employer: VMWare Inc (BROADCOM, after acquisition), USA – June 2022 to August 2024 **Senior Research and Development Engineer (Golang/Python)**

Project: Finch

(vSphere client plugin to deploy Tanzu)

Description: The project is built to demonstrate the end-to-end workflow to go from ESXi hosts to PAIF by radically simplifying the deployment of VCF and GPU accelerated Workload Domain. Enabling simplified on-premises deployments of VPAIF using automation across the VMware product stack and thereby providing a frictionless path for customers to realize value from their VCF subscriptions.

Software: **Golang, Python**

Environment: MacOS/Linux

Responsibilities:

- Implement Angular code(typescript) from scratch to consider inputs to create Tanzu cluster.
- Investigate and test REST APIs required to create Tanzu cluster with Postman.

- Implement proxy in Golang to route API requests to vCenter from angular code in POC.
- Implement session ID's that are required for plugin to connect to vCenter using GovMomi library.
- Test the application by validating the inputs and end to end workflow to create Tanzu cluster.

Project: IPV4 to IPV6 Translation

Description:

Implement Common appliance platform appliance that does translation from IPV4 to IPV6 using Jool Modules.

Software: **Golang**

Environment: Linux.

Responsibilities:

- Implement appliance from scratch and deploy in vRA.
- Test the application by writing unit tests with **Testify** Framework with the provided tools for assertions and mocks.
- Deploy plugin application as Google Cloud Function in **GCP** for the actual deployment of appliance.
- Worked on container management using **Docker** by writing Docker files.
- Used **Kubernetes** to orchestrate the deployment, scaling and management of docker containers
- Provide configuration for the appliance through frontend.

3) Employer: ERP Cloud Technologies, USA - March 2019 to May 2022

Client: Capital One, Texas (Contract)

Senior Golang Developer

Project: Managed Secrets Platform and Chamber of Secrets

Description:

Implementation of Managed secrets platform for All Access for rotation of passwords for Blue Green users for LDAP Active Directory, PostgreSQL, MySQL.

Software: **GoLang, AWS**

Environment: MacOS.

Responsibilities:

- Requirement's specifications and statistical analysis for new features.
- Implement AWS Lambdas in Golang to rotate passwords by connecting to LDAP URL and connecting to AWS RDS PostgreSQL.
- Work with Hashicorp vault and store passwords in lockbox paths created for specific user.
- Work on **AWS Lambda** in Golang to rotate ssh keys and update last rotation timestamps in DynamoDB for LDAP users.
- Responsible for setting up databases in AWS using RDS, storage using S3 buckets and configuring instance backups to S3 bucket.
- Create and manage **terraform** to automate various infrastructure tasks in AWS cloud infrastructure and deploy Microservices developed with Golang.
- Used Helm automating the Kubernetes applications lifecycle in a simple and consistent way.
- Used Jenkins pipelines to drive all micro services builds out to the **Docker registry** and then deployed to Kubernetes, created pods and managed using **Kubernetes**.
- Work on Multithreaded application using Go routines.
- Work on unit test **Go test** and mock libraries for testing the code from developer point of view.
- Work on writing test cases with Cucumber BDD **Godog** for API testing in Managed secrets Platform.

4) IBM (Systems & Technology Group), INDIA - June 2012 to Jan 2019

Senior Staff Software Engineer

Project: RSCT (IBM proprietary clustering product)

Description: Reliable Scalable Cluster Technology (RSCT) is a set of software components that together provide a comprehensive clustering environment for AIX, Linux, and Solaris. RSCT is infrastructure used by a variety of IBM products to provide clusters with improved system availability, scalability, and ease of use. Exploiters like HMC, VIOS, PowerHA use RSCT as internal product for high availability and clustering services.

Software: **C++, Python**

Environment: IBM AIX, RHEL 7, SLES 12 SP3, UBUNTU 17.10

Responsibilities:

- Requirements Analysis for new enhancements
- Developing, Design and unit testing of middleware and server side
- Developed multi-threaded, and multi-process and business logic using C++
- Took active part in Server-Side Development and coding using C++/STL/Unix.
- Configured GDB (GNU debugger) to analyze code flow, memory leak issues and core dump scenarios in the application.
- Worked on StorageRM and Topology services for heartbeat using TCP/IP Socket communication and which, on some cluster configurations, provides node/network failure detection.
- Worked on diagram tool which will call out any general TCP/IP network communication issues between the HMC and RMC of partition of KVM.
- Writing Python2 command line interface in IBM RSCT application.
- Building very good relationship with the team, customer and management.
- Developing new utility application support tools. Implementation and Unit testing.
- Requirements Analysis for new enhancements and fixing of issues.
- Recommending solutions and document findings.
- Used IPC for inter process communication
- Preparation of Unit test cases using CPP Unit Framework.
- Other miscellaneous tasks relating to the development and maintenance of a high-quality software product.
- Read /Write the different protocols.
- Porting of application from AIX to Linux (RHEL 6* and 7*), Ubuntu 17.10

5) ProArch IT Solutions - C, Python Developer, INDIA June 2010 - May 2012

Client: SEDC, Atlanta

Project: SMSListener (client-server application)

Description: The goal of the Project is to develop a SEDC Messaging System, which is a basic TCP sockets level interface, implements the standard sockets API calls and functionality and acts as an Interface between C# and COBOL. The server-side server program, SMSListner, is used to accept CSF client connections to the HP-UX computer through sockets and interfaces directly with SEDC's COBOL programs through Message Queues.

Software: C

Environment: HP-UX

Responsibilities:

- Played a major role in preparation of SRS.
- Played a major role in Designing of SMS Listener product in C/C++.
- Responsible for end-to-end TCP/IP socket communication between client (C#) and server(C) using IPC.
- Responsible for activating Micro Focus COBOL programs from SMS listener using UNIX system calls.
- Implemented two different modes, Persistent Mode and Standby mode for communication between SMSListener and COBOL programs using Multithreading.
- Responsible for Testing and Bug Fixing the code.
- Implemented logging with multiple levels and Error handling and coding in Shell.
- Have implemented a tool in client side for Load testing of server program.
- Responsible for Implementation of user interface (SMSMonitor) for SMSListener.
- Responsible for porting code to Redhat Linux.

6) Larsen and Toubro Infotech - C/C++ Developer, INDIA Jan 2007 - May 2010

Project (In-house): Infrastructure Management Systems Tools - May 2008 - May 2010

Description: To develop an Infrastructure Management product with Server Management Components to manage corporate infrastructure. The server management system will be developed as a component of the infrastructure management system, for managing the server parameters. The development of the server management system is with the aim of providing a cost effective and easily customizable solution to customers. The implementation will cover "OS monitoring" functionality to monitor various parameters of OS such as CPU, Memory and Disk Utilization etc.

Software: C, C++, STL.

Environment: Windows, SNMP Protocol and Win32 API'S

Responsibilities:

- Coding the programs in accordance with Program Specifications and adhering to Coding Standards.
- Involved in the preparation of SRS, HLD and DLD.
- Worked in C/C++ with SNMP Protocol and WIN32 API's for retrieving OS Parameters such as CPU, Memory, Disk space etc.
- Participating in code review activities.
- Bug Fixing in the code.

Project: 3G-LTE/WIMAX Simulator Jan 2007 - April 2008

Client: SAMSUNG

Description: LTE simulator simulates 3G-LTE Physical Layer at both sides downlink (OFDMA) as well as uplink (SCFDMA) based on 3GPP Technical Specifications release 8. LTE simulator is powerful, PC-based software for creating standards-based LTE signals. An easy-to-use user interface simplifies signal creation. These signals help engineers better characterize evaluate and fine-tune their designs under parametric and functional test conditions. The software features the ability to adjust LTE parameters to fine-tune signal characteristics for specific applications. Some of the key parameters that can be adjusted include bandwidth (1.25-, 2.5-, 5-, 10-, 15- or 20MHz), modulation type (QPSK, 16QAM or 64QAM), number of channels, type of channels (such as data or control), channel power level, payload data and resource block allocation.

Software: C, C++, Multithreading, STL.

Data Base: SQL

Responsibilities:

- Coding and implementation of the entire solution for data acquisition
- Review of design document, code and test case documents.
- The implementation was done, keeping in mind the memory and speed constraints while developing the algorithms.
- Used C++/ STL as containers for Database Objects.
- Performed development on both windows and Unix environment.
- Used SQL as Database and involved in writing the SQL queries.
- Conducted super bench and cell integration testing.
- Debugged and fixed various issues which were caused due to the porting of code.
- Porting of application from Windows to Linux, HP-UX, Sun Solaris.
- Used SVN as Version Control System for managing the Application Code.